

MOHAMED ALI BENBELGHITH

Software Engineering Student · ML Engineer · Open Source Contributor

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PROFILE

Software Engineering student at INSAT (GL2) specializing in machine learning, computer vision, and AI systems. Active open-source contributor (SunPy, ImageLab, TensorMap, DataLoom) with GSoC 2026 proposals across three organizations. Strong foundations in topological data analysis and full-stack development (Python, Java/Symfony, TypeScript).

EDUCATION

Software Engineering (Génie Logiciel — GL2) · INSAT, University of Carthage, Tunisia · 2024 – Present

- Linear Algebra, Databases, System Administration, Network Programming
- UML modeling & enterprise architecture (Enterprise Architect)

GOOGLE SUMMER OF CODE 2026 — PROPOSALS

RadioSpectra Modernization · SunPy / OpenAstronomy

- Designed WCS/NDCube integration for improved coordinate-system handling in radio spectrogram data
- Proposed advanced background subtraction algorithms (SNR-based, polynomial baseline, morphological)
- Engineered gap-aware visualization with dynamic spectrum rendering for multi-instrument data

Interactive blog: mohamedali1937.github.io/sunpy_blog

ImageLab Pipeline Enhancement · C2SI / ImageLab

- Designed per-step preview functionality and histogram visualization for image pipelines
- Proposed a Blockly + FastAPI + React + OpenCV visual programming architecture for no-code CV pipelines
- Outlined a macro operator system for complex, reusable transformation workflows
- Developed pipeline persistence and batch processing for production-grade workflows

DataLoom Framework · C2SI / DataLoom

- Comprehensive proposal for data processing and transformation framework enhancements
- Outlined technical architecture improvements and new feature implementations

TensorMap Visual ML Platform · C2SI / TensorMap

- Contributed to a visual programming platform for designing ML algorithms without writing code
- Proposed enhancements to neural network architecture design, layer configuration, and TF code generation
- Aimed at lowering the barrier for beginners via an intuitive drag-and-drop interface

OPEN SOURCE CONTRIBUTIONS

- Active contributor across C2SI and SunPy ecosystems — forks maintained at github.com/MohamedAli1937

GSoC proposals archive with full technical documentation: github.com/MohamedAli1937/gsoc-proposals

FEATURED PROJECTS

GitFlow Dashboard | [TypeScript](#) · [Git](#) · [DevTools](#)

An interactive dashboard for visualizing Git branching workflows across feature, release, and hotfix flows. Renders branch timelines, merge histories, and diff summaries to help developers track project pipelines and improve team productivity.

TDA Market Risk Analyzer | [Mathematics](#) · [TDA](#) · [Python](#)

A real-time financial risk detection engine built on Topological Data Analysis, using Takens' embedding and Vietoris–Rips complexes to surface structural shifts in market data ahead of traditional indicators. Wrote an accompanying blog breaking down the TDA methodology and its application to financial risk.

Interactive blog: mohamedali1937.github.io/tda_blog

Certificate DApp | [Solidity](#) · [Ethereum](#) · [React](#)

A decentralized application for issuing and verifying academic certificates on the Ethereum blockchain. Integrates MetaMask for wallet authentication and uses Solidity smart contracts for tamper-proof, on-chain verification of credentials.

MITRE ATLAS Knowledge Graph | [Python](#) · [Knowledge Graphs](#) · [AI Security](#)

A knowledge graph system mapping adversarial machine learning threats using the MITRE ATLAS framework, structuring tactics, techniques, and case studies to support AI security research and threat modeling.

Alzheimer's MRI Prediction | [TensorFlow](#) · [Keras](#) · [Python](#)

A deep learning pipeline for early-stage Alzheimer's detection from brain MRI scans, using CNNs with transfer learning, data augmentation, and multi-class classification to demonstrate real-world clinical diagnostic applicability.

KIBRA Smart Packing | [TypeScript](#) · [Gemini Pro](#) · [AI](#)

An AI-powered logistics optimization system that leverages the Google Gemini 3 PRO API for intelligent 3D packing decisions, built on a sophisticated TypeScript architecture with constraint-based reasoning for space-efficient loading.

Sign Language Detection | [OpenCV](#) · [MediaPipe](#) · [Python](#)

A real-time American Sign Language recognition system using hand landmark tracking and gesture classification, built as accessible technology to support communication for hearing-impaired users.

Sentiment Analyzer | [Python](#) · [NLTK](#) · [Scikit-learn](#)

A full-stack NLP web platform performing text preprocessing, feature extraction, and multi-class sentiment classification, exposed through a Flask backend for analyzing emotional tone in text data.

Real-time Face Detection | [Python](#) · [OpenCV](#) · [Dlib](#)

A high-speed computer vision system for facial landmark tracking and identification, optimized for low-latency detection in live video streams.

Cpp-Shell | [C++](#) · [Systems Programming](#)

A custom UNIX-like shell interpreter built in C++ from scratch, implementing built-in commands, piping, I/O redirection, and process management to demonstrate core operating systems concepts.

Desert Journey Portfolio | [Web](#) · [Interactive Design](#)

An immersive, narrative-driven interactive web portfolio that guides visitors on a desert-journey theme, showcasing projects and skills in a visually distinctive storytelling format.

TECHNICAL SKILLS

- Languages: Python, TypeScript, JavaScript, Java, PHP, C/C++, SQL, Solidity
- ML/DL & CV: TensorFlow, PyTorch, Keras, Scikit-learn, CNNs, OpenCV, MediaPipe, TDA
- Web & Backend: Symfony, Flask, FastAPI, React, Node.js, REST APIs
- AI & APIs: Gemini API, LangChain, Anthropic API, Blockly
- Data: Pandas, NumPy, Matplotlib, Jupyter | DevOps: Git, Docker, Linux, CI/CD
- Databases: MySQL, PostgreSQL, Oracle, Doctrine ORM | Blockchain: Ethereum, Solidity, Web3.js

ACHIEVEMENTS

- GSoC 2026 proposals — RadioSpectra, ImageLab, TensorMap, DataLoom
- GitHub Pull Shark — merged PRs across multiple open-source repos
- Published interactive TDA blog for financial & scientific data visualization